

Explicit Zero-Free Regions for Dirichlet L -Functions

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Let $\mathcal{L}_k(s)$ be the product of the $\varphi(k)$ Dirichlet L -functions formed with characters modulo k . We prove the existence of explicit numerical zero-free regions for $\mathcal{L}_k(s)$. The first result is that $\mathcal{L}_k(s)$ has at most a single zero in the region $\{s: \sigma > 1 - 1/(R \log M)\}$, where $R = 9.645908801$ and $M = \max\{k, k|t|, 10\}$. The only possible zero in this region is a simple real zero arising from an L -function formed with a real non-principal character. The second result is that if χ_1 and χ_2 are distinct real primitive characters modulo k_1 and k_2 , respectively, and if β_i is a zero of $L(s, \chi_i)$, $i = 1, 2$, then $\min\{\beta_1, \beta_2\} < 1 - 1/(R_1 \log M_1)$, where $R_1 = (5 - \sqrt{5})/(15 - 10\sqrt{2})$, and $M_1 = \max\{k_1 k_2/17, 13\}$.

1. INTRODUCTION

Let k be a positive integer, $s = \sigma + it$ be a complex variable, and $\mathcal{L}_k(s)$ be the product of the $\varphi(k)$ Dirichlet L -functions formed with characters modulo k . The main purpose of this paper is to prove the existence of an explicit region containing at most a single zero of $\mathcal{L}_k(s)$. We also prove two results concerning the possible existence of a zero within this region.

In the case $k = 1$ we have $\mathcal{L}_1(s) = \zeta(s)$, the Riemann zeta function. A classical result due to de la Vallée Poussin [17] states that there exists a constant C_1 such that $\zeta(s) \neq 0$ for

$$\sigma \geq 1 - \frac{C_1}{\log |t|}, \quad |t| \geq t_0. \quad (1)$$

As a consequence of this he was able to prove that there exists a constant C_2 such that the error term in the prime number theorem satisfies

$$\psi(x) - x = O(xe^{-C_2 \sqrt{\log x}}). \quad (2)$$

Various estimates of the constants C_1 , t_0 , and C_2 have been given by de la Vallée Poussin, Landau [4], Rosser and Schoenfeld [11], and Stechkin [14].

The best known result for C_1 is that of Rosser and Schoenfeld [12], who gave the value $C_1 = 1/R$, where $R = 9.645908801$. From this Schoenfeld [13] was able to prove that

$$|\psi(x) - x| < x \sqrt{\frac{8}{17\pi}} \left(\frac{\log x}{R}\right)^{1/4} \exp\left(-\sqrt{\frac{\log x}{R}}\right), \quad x \geq 17. \quad (3)$$

In 1899 de la Vallée Poussin [17] stated that it was possible to prove (1) by the same technique in the case $k > 1$. From this he deduced that the error term in the prime number theorem for arithmetic progressions satisfies

$$\psi(x; k, l) - \frac{x}{\phi(k)} = O(x e^{-C_2 \sqrt{\log x}}), \quad (4)$$

provided $(k, l) = 1$. This is indeed the case for a fixed value of k , but the constants C_1, C_2 , and the implied O -constant will depend on k , and de la Vallée Poussin gave no indication as to the nature of this dependence.

In many applications it is desirable to have an estimate such as (4) hold uniformly in a certain range of k relative to x . This requires the knowledge of a zero free region for $\mathcal{L}_k(s)$ that is uniform in k , and work in this direction was carried out by Gronwall [3], Titchmarsh [15], Page [8], and Landau [5]. To a great extent this paper may be regarded as an up-to-date version of their work. The greatest difficulty lies in the estimation of the real zeros of L -functions formed with real non-principal characters modulo k . For these zeros the best known estimates are considerably weaker than in all other cases. This is reflected in the following result.

THEOREM 1. *Let $M = \max\{k, k|t|, 10\}$ and $R = 9.645908801$. Then $\mathcal{L}_k(s)$ has at most a single zero in the region*

$$\{s : \sigma \geq 1 - 1/(R \log M)\}.$$

The only possible zero in this region is a simple real zero arising from an L -function formed with a real non-principal character modulo k .

If a real zero β of $\mathcal{L}_k(s)$ exists for which $\beta > 1 - 1/(R \log k)$, then we shall refer to β as an exceptional zero, and k as an exceptional modulus. Non-exceptional moduli are of particular interest since for these k Theorem 1 allows us to prove sharp explicit numerical bounds for the Chebyshev functions $\psi(x; k, l)$ and $\theta(x; k, l)$. This will be the subject of a later paper.

The following result can be used to show that certain moduli are not exceptional.

THEOREM 2. *Let χ_1 and χ_2 be distinct real primitive characters modulo k_1 and k_2 , respectively, and let β_i be a real zero of $L(s, \chi_i)$, $i = 1, 2$. Let $M_1 = \max\{k_1 k_2 / 17, 13\}$ and let $R_1 = (5 - \sqrt{5}) / (15 - 10\sqrt{2})$. Then*

$$\min\{\beta_1, \beta_2\} < 1 - \frac{1}{R_1 \log M_1}.$$

Theorem 2 was first proved by Landau with an unspecified constant in place of R_1 . As a consequence of Theorem 2 we may conclude that, under certain conditions on the multiplicative structure of two moduli, at most one of the two moduli can be exceptional. For a simple example, we have the following result.

COROLLARY. *If k_1 and k_2 are relatively prime integers with $k_1 < k_2 \leq k_1^{(R/R_1)-1}$, then at most one of k_1 or k_2 is an exceptional modulus.*

Rosser has shown [9, 10] that if χ is a real primitive character modulo k with $k \leq 227$, then $L(s, \chi)$ has no positive real zeros. He has also extended this to $k \leq 986$ in an unpublished manuscript. Thus if an exceptional modulus exists it must exceed 986. Furthermore, it is well known that real primitive characters exist only for moduli of the form $2^a Q$, where $a = 0, 2$, or 3 and Q is an odd squarefree integer. Hence from Rosser's work it is evident that if k is odd and the squarefree kernel of k does not exceed 986, then k is not exceptional. Similar results hold for even moduli.

2. NOTATION AND LEMMAS

The method of proof for Theorem 1 has its origins in the work of de la Vallée Poussin, but we incorporate many refinements due to Gronwall, Titchmarsh, Landau, Page, Rosser and Schoenfeld, Metsänkylä [7], and the author [6]. The method of de la Vallée Poussin involves the use of a non-negative cosine polynomial. In several cases we shall use the polynomial

$$P(\theta) = 8(0.9126 + \cos \theta)^2(0.2766 + \cos \theta)^2.$$

This polynomial is the same as that used by Rosser and Schoenfeld [12] for $\zeta(s)$, and is nearly optimal for our purposes. We write

$$P(\theta) = \sum_{m=0}^4 a_m \cos m\theta,$$

where

$$a_0 = 11.1859355312082048,$$

$$a_1 = 19.073344004352,$$

$$a_2 = 11.67618784,$$

$$a_3 = 4.7568,$$

$$a_4 = 1.$$

For simplicity we write

$$\begin{aligned} A &= \sum_{m=1}^4 a_m \\ &= 36.506331844352. \end{aligned}$$

The parameter σ is to be restricted so that $1 < \sigma < 1.3$, and in some cases we shall impose the further restriction that $\sigma < 1.15$. The parameter σ_1 is defined by

$$\sigma_1 = \frac{1}{2} + \frac{1}{2} \sqrt{1 + 4\sigma^2}.$$

For real t we define

$$\begin{aligned} f(t, \chi) &= f(\sigma, t, \chi) \\ &= \operatorname{Re} \left[\frac{1}{\sqrt{5}} \frac{L'}{L} (\sigma_1 + it, \chi) - \frac{L'}{L} (\sigma + it, \chi) \right]. \end{aligned} \quad (5)$$

Finally we define

$$\begin{aligned} T(k, k_1) &= \sum_{\substack{p|k \\ p \nmid k_1}} (\log p) \left(\frac{1}{p^\sigma - 1} + \frac{1}{\sqrt{5} (p^{\sigma_1} - 1)} \right), \\ s(k) &= \sum_{p|k} (\log p) \left(\frac{1}{p^\sigma - 1} - \frac{1}{\sqrt{5} (p^{\sigma_1} - 1)} \right), \\ \kappa &= (5 - \sqrt{5})/10, \\ \phi &= (1 + \sqrt{5})/2, \\ F(s, z) &= \operatorname{Re} \left[\frac{1}{s - z} + \frac{1}{s - 1 + \bar{z}} \right]. \end{aligned}$$

LEMMA 1. Let $1 < \sigma < 1.15$, $|t| < 1$, $1 \leq m \leq 4$, $0 \leq a \leq 2$, $s = \sigma + imt$, and $s_1 = \sigma_1 + imt$. Then

$$\frac{1}{2} \operatorname{Re} \left[\frac{\Gamma'}{\Gamma} \left(\frac{s+a}{2} \right) - \frac{1}{\sqrt{5}} \frac{\Gamma'}{\Gamma} \left(\frac{s_1+a}{2} \right) \right] < d(a, m),$$

where $d(a, m)$ is given by the following table.

m	a	
	0 or 1	2
1	-0.0390	0.0615
2	0.2469	0.1565
3	0.4452	0.2638
4	0.5842	0.3636

Proof. Let $z = x + iy$, with $x > 0$. Then

$$\frac{\Gamma'}{\Gamma}(z) = -\gamma - \frac{1}{z} + \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+z} \right), \quad (6)$$

so that $\operatorname{Re}(\Gamma'(z)/\Gamma(z))$ is clearly an increasing function of $|y|$. Hence we have

$$\begin{aligned} & \frac{1}{2} \operatorname{Re} \left[\frac{\Gamma'}{\Gamma} \left(\frac{s+a}{2} \right) - \frac{1}{\sqrt{5}} \frac{\Gamma'}{\Gamma} \left(\frac{s_1+a}{2} \right) \right] \\ & < \frac{1}{2} \operatorname{Re} \frac{\Gamma'}{\Gamma} \left(\frac{\sigma+a+im}{2} \right) - \frac{1}{2\sqrt{5}} \frac{\Gamma'}{\Gamma} \left(\frac{\sigma_1+a}{2} \right). \end{aligned} \quad (7)$$

It follows from (6) that

$$\begin{aligned} \operatorname{Re} \frac{\Gamma'}{\Gamma} \left(\frac{\sigma+a+im}{2} \right) & < -\gamma - \frac{2(1+a)}{(1.15+a)^2 + m^2} \\ & + \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{2(1+a) + 4n}{(1.15+a+2n)^2 + m^2} \right). \end{aligned} \quad (8)$$

It is a simple matter to estimate the error made in truncating the sum after N terms. For example, if $a = 2$ then we have

$$\begin{aligned}
\sum_{n=N+1}^{\infty} \left(\frac{1}{n} - \frac{4n+6}{(3.15+2n)^2+m^2} \right) &< \sum_{n=N+1}^{\infty} \frac{25.9225+6.6n}{n[(3.15+2n)^2+16]} \\
&< \sum_{n=N+1}^{\infty} \frac{2}{n^2} \\
&< \frac{2}{N}.
\end{aligned}$$

The cases $a = 0$ or 1 are similar and yield the same estimate.

The sums in (8) were computed using 1000 terms, and were then rounded up and increased by 0.002 to account for the above error estimate.

In order to complete the proof, note that

$$\frac{\Gamma'}{\Gamma} \left(\frac{\sigma_1 + a}{2} \right) > \frac{\Gamma'}{\Gamma} (0.805 + a/2),$$

and these values can be found in Abramowitz and Stegun [1].

LEMMA 2. *Let $1 < \sigma < 1.15$, $t > 1$, $1 \leq m \leq 4$, $0 \leq a \leq 2$, $s = \sigma + imt$, and $s_1 = \sigma_1 + imt$. Then*

$$\frac{1}{2} \operatorname{Re} \left[\frac{\Gamma'}{\Gamma} \left(\frac{s+a}{2} \right) - \frac{1}{\sqrt{5}} \frac{\Gamma'}{\Gamma} \left(\frac{s_1+a}{2} \right) \right] < \kappa \log t + c(a, m),$$

where $\kappa = (5 - \sqrt{5})/10$ and $c(a, m)$ is given by the following table.

m	a	
	0 or 1	2
1	0.3918	0.3316
2	0.3915	0.3530
3	0.4062	0.3780
4	0.4266	0.4080

Proof. Let $z = x + iy$, with $x > 0$. Then

$$\begin{aligned}
\operatorname{Re} \frac{\Gamma'}{\Gamma} \left(\frac{z+a}{2} \right) &= \log \left| \frac{z+a}{2} \right| - \frac{x+a}{(x+a)^2 + y^2} \\
&\quad + \operatorname{Re} \int_0^\infty \frac{u - [u] - 1/2}{(u + (z+a)/2)^2} du.
\end{aligned}$$

The last term satisfies

$$\begin{aligned} & \left| \operatorname{Re} \int_0^\infty \frac{u - [u] - 1/2}{(u + (z + a)/2)^2} du \right| \\ & < \frac{1}{2} \int_0^\infty \frac{du}{|u + (z + a)/2|^2} = \frac{1}{y} \left[\frac{\pi}{2} - \tan^{-1} \left(\frac{x + a}{y} \right) \right], \end{aligned}$$

and if $y \geq m$ it follows that

$$\begin{aligned} \operatorname{Re} \frac{\Gamma'}{\Gamma} \left(\frac{z + a}{2} \right) &= \log \frac{y}{2} + \frac{1}{2} \log \left| 1 + \left(\frac{x + a}{y} \right)^2 \right| \\ &\quad - \frac{x + a}{(x + a)^2 + y^2} + \frac{\theta}{m} \left[\frac{\pi}{2} - \tan^{-1} \left(\frac{x + a}{m} \right) \right], \end{aligned}$$

where $|\theta| \leq 1$. We subtract to obtain

$$\begin{aligned} & \frac{1}{2} \operatorname{Re} \left[\frac{\Gamma'}{\Gamma} \left(\frac{s + a}{2} \right) - \frac{1}{\sqrt{5}} \frac{\Gamma'}{\Gamma} \left(\frac{s_1 + a}{2} \right) \right] \\ &= \kappa \log \frac{mt}{2} + \frac{1}{4} \log \left| 1 + \left(\frac{\sigma + a}{mt} \right)^2 \right| \\ &\quad - \frac{1}{4\sqrt{5}} \log \left[1 + \left(\frac{\sigma_1 + a}{mt} \right)^2 \right] \\ &\quad - \frac{1}{2} \frac{\sigma + a}{(\sigma + a)^2 + m^2 t^2} + \frac{1}{2\sqrt{5}} \frac{\sigma_1 + a}{(\sigma_1 + a)^2 + m^2 t^2} \\ &\quad + \frac{\theta_1}{2m} \left[\frac{\pi}{2} - \tan^{-1} \left(\frac{1 + a}{m} \right) \right] + \frac{\theta_2}{2m\sqrt{5}} \left[\frac{\pi}{2} - \tan^{-1} \left(\frac{\phi + a}{m} \right) \right], \end{aligned} \tag{9}$$

where $|\theta_i| \leq 1$. It is easily verified that for all $t > 0$ we have

$$-\frac{1}{2} \frac{\sigma + a}{(\sigma + a)^2 + m^2 t^2} + \frac{1}{2\sqrt{5}} \frac{\sigma_1 + a}{(\sigma_1 + a)^2 + m^2 t^2} < 0. \tag{10}$$

Furthermore we have

$$\begin{aligned} & \frac{1}{4} \log \left[1 + \left(\frac{\sigma + a}{mt} \right)^2 \right] - \frac{1}{4\sqrt{5}} \log \left[1 + \left(\frac{\sigma_1 + a}{mt} \right)^2 \right] \\ & < \frac{1}{4} \log \left[1 + \left(\frac{1.15 + a}{mt} \right)^2 \right] - \frac{1}{4\sqrt{5}} \log \left[1 + \left(\frac{\phi + a}{mt} \right)^2 \right], \end{aligned}$$

and the latter is a decreasing function of t . It follows from (9) and (10) that

$$\begin{aligned} \frac{1}{2} \operatorname{Re} \left[\frac{\Gamma'}{\Gamma} \left(\frac{s+a}{2} \right) - \frac{1}{\sqrt{5}} \frac{\Gamma'}{\Gamma} \left(\frac{s_1+a}{2} \right) \right] &< \kappa \log t + \kappa \log \frac{m}{2} \\ &+ \frac{1}{4} \log \left[1 + \left(\frac{1.15+a}{m} \right)^2 \right] - \frac{1}{4\sqrt{5}} \log \left[1 + \left(\frac{\phi+a}{m} \right)^2 \right] \\ &+ \frac{1}{2m} \left[\frac{\pi}{2} - \tan^{-1} \left(\frac{1+a}{m} \right) \right] + \frac{1}{2m\sqrt{5}} \left[\frac{\pi}{2} - \tan^{-1} \left(\frac{\phi+a}{m} \right) \right]. \end{aligned}$$

These values were computed and rounded up, using the larger of the two values for $a = 0$ or 1 .

LEMMA 3. *Let χ_0 be the principal character modulo k . If $1 < \sigma < 1.15$, then*

$$f(0, \chi_0) < \frac{1}{\sigma - 1} - 0.8973 - s(k).$$

If $1 < \sigma < 1.3$, then

$$f(0, \chi_0) < \frac{1}{\sigma - 1} - 0.7833.$$

Proof. It follows from (5) that

$$f(0, \chi_0) = \frac{1}{\sqrt{5}} \frac{\zeta'}{\zeta}(\sigma_1) - \frac{\zeta'}{\zeta}(\sigma) - s(k). \quad (11)$$

In order to estimate $\zeta'(\sigma_1)/\zeta(\sigma_1)$, note that

$$\frac{\zeta'}{\zeta}(\sigma_1) = - \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{\sigma_1}} = - \sum_p \frac{\log p}{p^{\sigma_1} - 1}.$$

This is evidently an increasing function of σ_1 , and we achieve the largest value when $\sigma = 1.15$ or 1.3 , respectively. From the terms with $p < 250$ we get

$$\begin{aligned} \frac{1}{\sqrt{5}} \frac{\zeta'}{\zeta}(\sigma_1) &< -0.3783, & 1 < \sigma < 1.15, \\ &< -0.2993, & 1 < \sigma < 1.3. \end{aligned} \quad (12)$$

From Davenport [2] we have

$$-\frac{\zeta'}{\zeta}(\sigma) = \frac{1}{\sigma-1} - \frac{1}{2} \log \pi + \frac{1}{2} \frac{\Gamma'}{\Gamma} \left(\frac{\sigma+2}{2} \right) - \sum_{\rho} \frac{\sigma-\beta}{(\sigma-\beta)^2 + \gamma^2},$$

where the sum on ρ extends over all zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ in the critical strip. The graph of the function $\Gamma'(z)/\Gamma(z)$ is concave down for $z > 0$, so that

$$\begin{aligned} \frac{\Gamma'}{\Gamma} \left(\frac{\sigma+2}{2} \right) &< \frac{\Gamma'}{\Gamma} \left(\frac{3}{2} \right) + \frac{d}{d\sigma} \left(\frac{\Gamma'}{\Gamma} \left(\frac{\sigma+2}{2} \right) \right)_{\sigma=1} (\sigma-1) \\ &= 2 - C - \log 4 + \left(\frac{\pi^2}{4} - 2 \right) (\sigma-1), \end{aligned}$$

where C is Euler's constant. It follows that

$$-\frac{\zeta'}{\zeta}(\sigma) < \frac{1}{\sigma-1} - \frac{1}{2} \log \pi + 1 - \frac{C}{2} - \log 2 + \left(\frac{\pi^2}{8} - 1 \right) (\sigma-1),$$

since the sum on ρ is non-negative.

The result then follows from (11), (12), and the fact that $s(k) \geq 0$.

LEMMA 4 (Stechkin). *Let $\sigma > 1$, $\sigma_1 = \frac{1}{2} + \frac{1}{2} \sqrt{1+4\sigma^2}$, $s = \sigma + it$, $s_1 = \sigma_1 + it$, and $F(s, z) = \operatorname{Re}[1/(s-z) + 1/(s-1+\bar{z})]$.*

If $0 < \operatorname{Re} z < 1$, then

$$F(s, z) - \frac{1}{\sqrt{5}} F(s_1, z) \geq 0.$$

If $\operatorname{Im} z = t$ and $\frac{1}{2} \leq \operatorname{Re} z < 1$, then

$$\operatorname{Re} \frac{1}{s-1+\bar{z}} - \frac{1}{\sqrt{5}} F(s_1, z) \geq 0.$$

Proof. See Stechkin [14].

LEMMA 5. *Let χ be a primitive non-principal character modulo k , $1 < \sigma < 1.15$, and let $\rho = \beta + i\gamma$ be a zero of $L(s, \chi)$ with $\beta \geq \frac{1}{2}$.*

If $0 \leq \gamma < 1$, then

$$f(\gamma, \chi) < \kappa \log k + d(1, 1) - \kappa \log \pi - \frac{1}{\sigma - \beta}.$$

If $\gamma \geq 1$, then

$$f(\gamma, \chi) < \kappa \log k\gamma + c(1, 1) - \kappa \log \pi - \frac{1}{\sigma - \beta}.$$

Proof. From Davenport [2] we have

$$-\operatorname{Re} \frac{L'}{L}(s, \chi) = \frac{1}{2} \log \frac{k}{\pi} + \frac{1}{2} \operatorname{Re} \frac{\Gamma'}{\Gamma} \left(\frac{s+a}{2} \right) - \sum_{\beta \geq 1/2}' F(s, \rho)$$

where the dash “'” indicates that the terms of the sum with $\beta = \frac{1}{2}$ are to be counted with weight $\frac{1}{2}$, and $a = (1 - \chi(-1))/2$. Subtracting we get

$$\begin{aligned} f(t, \chi) &= \kappa \log \frac{k}{\pi} + \frac{1}{2} \operatorname{Re} \left[\frac{\Gamma'}{\Gamma} \left(\frac{s+a}{2} \right) - \frac{1}{\sqrt{5}} \frac{\Gamma'}{\Gamma} \left(\frac{s_1+a}{2} \right) \right] \\ &\quad - \sum_{\beta \geq 1/2}' \left[F(s, \rho) - \frac{1}{\sqrt{5}} F(s_1, \rho) \right]. \end{aligned} \quad (13)$$

By Lemma 4 the terms of the last sum are non-negative. We discard all terms except for the one corresponding to the zero of interest, and set $t = \gamma$ to get

$$\begin{aligned} f(\gamma, \chi) &< \kappa \log \frac{k}{\pi} + \frac{1}{2} \operatorname{Re} \left[\frac{\Gamma'}{\Gamma} \left(\frac{s+a}{2} \right) - \frac{1}{\sqrt{5}} \frac{\Gamma'}{\Gamma} \left(\frac{s_1+a}{2} \right) \right] \\ &\quad - \frac{1}{\sigma - \beta} - \frac{1}{\sigma - 1 + \beta} + \frac{1}{\sqrt{5}} F(\sigma_1, \beta). \end{aligned} \quad (14)$$

If $0 \leq \gamma < 1$ then the result follows from (14), Lemma 1, and Lemma 4. If $\gamma \geq 1$ then we use Lemma 2 in place of Lemma 1.

LEMMA 6. Let χ be a non-principal character modulo k , $1 < \sigma < 1.15$, and let m be an integer with $1 \leq m \leq 4$.

If χ is primitive, then

$$\begin{aligned} f(mt, \chi) &< \kappa \log k + d(1, m) - \kappa \log \pi, & 0 \leq t < 1, \\ &< \kappa \log kt + c(1, m) - \kappa \log \pi, & t \geq 1. \end{aligned}$$

If χ is imprimitive with conductor k_1 , then

$$\begin{aligned} f(mt, \chi) &< \kappa \log k_1 + d(1, m) - \kappa \log \pi + T(k, k_1), & 0 \leq t < 1, \\ &< \kappa \log k_1 t + c(1, m) - \kappa \log \pi + T(k, k_1), & t \geq 1. \end{aligned}$$

Proof. Consider first the case where χ is primitive. In (13) we replace t by mt and use Lemma 4 to discard the sum on ρ . If $t \geq 1$ we use Lemma 2, and if $0 \leq t < 1$ we use Lemma 1. This completes the proof for χ primitive.

If χ is induced by the primitive character χ_1 modulo k_1 , then

$$\frac{L'}{L}(s, \chi) = \frac{L'}{L}(s, \chi_1) + \sum_{\substack{p \mid k \\ p \nmid k_1}} \frac{\chi_1(p) \log p}{p^s - 1}. \quad (15)$$

It follows from (5) and (15) that

$$f(mt, \chi) = f(mt, \chi_1) + \sum_{\substack{p \mid k \\ p \nmid k_1}} \operatorname{Re} \chi_1(p) \log p \left\{ \frac{1}{\sqrt{5}(p^{s_1} - 1)} - \frac{1}{p^s - 1} \right\}. \quad (16)$$

The result then follows from the first part of this lemma and the obvious estimate for the sum in (16).

LEMMA 7. Let χ_0 be the principal character modulo k , $1 < \sigma < 1.15$, and $2 \leq m \leq 4$. Then

$$\begin{aligned} f(mt, \chi_0) &< \frac{\sigma - 1}{(\sigma - 1)^2 + m^2 t^2} + d(2, m) - \kappa \log \pi + T(k, 1), & 0 \leq t < 1 \\ &< \kappa \log t + c(2, m) - \kappa \log \pi + T(k, 1), & t \geq 1. \end{aligned}$$

Proof. From Davenport [2] we have

$$\frac{L'}{L}(s, \chi_0) = \frac{\zeta'}{\zeta}(s) + \sum_{p \mid k} \frac{\log p}{p^s - 1},$$

and

$$\begin{aligned} -\operatorname{Re} \frac{\zeta'}{\zeta}(s) &= -\frac{1}{2} \log \pi + \operatorname{Re} \frac{1}{s - 1} + \frac{1}{2} \operatorname{Re} \frac{\Gamma'}{\Gamma} \left(\frac{s + 2}{2} \right) \\ &\quad - \sum_{\beta \geq 1/2} F(s, \rho), \end{aligned} \quad (17)$$

where the sum extends over the zeros of $\zeta(s)$. Thus we have

$$\begin{aligned} f(mt, \chi_0) &= \operatorname{Re} \left(\frac{1}{\sqrt{5}} \frac{\zeta'}{\zeta}(s_1) - \frac{\zeta'}{\zeta}(s) \right) \\ &\quad + \operatorname{Re} \sum_{p \mid k} \log p \left(\frac{1}{\sqrt{5}(p^{s_1} - 1)} - \frac{1}{p^s - 1} \right), \end{aligned} \quad (18)$$

and

$$\begin{aligned}
 & \operatorname{Re} \left(\frac{1}{\sqrt{5}} \frac{\zeta'}{\zeta}(s_1) - \frac{\zeta'}{\zeta}(s) \right) \\
 &= -\kappa \log \pi + \frac{1}{2} \operatorname{Re} \left[\frac{\Gamma'}{\Gamma} \left(\frac{s+2}{2} \right) - \frac{1}{\sqrt{5}} \frac{\Gamma'}{\Gamma} \left(\frac{s_1+2}{2} \right) \right] \\
 &+ \frac{\sigma-1}{(\sigma-1)^2 + m^2 t^2} - \frac{\sigma_1-1}{\sqrt{5} [(\sigma_1-1)^2 + m^2 t^2]} \\
 &- \sum'_{\beta \geq 1/2} \left[F(s, \rho) - \frac{1}{\sqrt{5}} F(s_1, \rho) \right]. \tag{19}
 \end{aligned}$$

If $t \geq 1$ then we have

$$\frac{\sigma-1}{(\sigma-1)^2 + m^2 t^2} - \frac{\sigma_1-1}{\sqrt{5} [(\sigma_1-1)^2 + m^2 t^2]} < 0. \tag{20}$$

The terms of the sum on ρ in (19) are non-negative by Lemma 4, so that the result follows for $t \geq 1$ from (18), (19), (20), and Lemma 2. The proof for $0 \leq t < 1$ uses Lemma 1 and the fact that $\sigma_1 - 1 > 0$.

LEMMA 8. *Let χ be a primitive non-principal character modulo k , and let $1 < \sigma < 1.3$. Then*

$$\begin{aligned}
 f(0, \chi) + f(0, \bar{\chi}) &< 2\kappa \log k - 0.91398 \\
 &- 2 \sum'_{\beta \geq 1/2} \left[F(\sigma, \rho) - \frac{1}{\sqrt{5}} F(\sigma_1, \rho) \right],
 \end{aligned}$$

where the sum extends over all zeros of $L(s, \chi)$ in the critical strip.

Proof. From (13) we obtain

$$\begin{aligned}
 f(0, \chi) + f(0, \bar{\chi}) &= 2\kappa \log \frac{k}{\pi} + \frac{\Gamma'}{\Gamma} \left(\frac{\sigma+a}{2} \right) - \frac{1}{\sqrt{5}} \frac{\Gamma'}{\Gamma} \left(\frac{\sigma_1+a}{2} \right) \\
 &- \sum'_{\beta \geq 1/2} \left[F(\sigma, \rho) - \frac{1}{\sqrt{5}} F(\sigma_1, \rho) \right],
 \end{aligned}$$

where the sum extends over all zeros of $L(s, \chi) L(s, \bar{\chi})$ counting multiplicities. Note that the zeros occur in complex conjugate pairs, since $L(\bar{\rho}, \bar{\chi}) = 0$ whenever $L(\rho, \chi) = 0$. Hence we can multiply the sum by 2 and include only zeros of $L(s, \chi)$.

The function $\Gamma'(x)/\Gamma(x)$ is increasing for positive x , so that if $1 < \sigma < 1.3$, then we have

$$\begin{aligned} \frac{\Gamma'}{\Gamma} \left(\frac{\sigma + a}{2} \right) - \frac{1}{\sqrt{5}} \frac{\Gamma'}{\Gamma} \left(\frac{\sigma_1 + a}{2} \right) \\ < \frac{\Gamma'}{\Gamma} \left(0.65 + \frac{a}{2} \right) - \frac{1}{\sqrt{5}} \frac{\Gamma'}{\Gamma} \left(0.805 + \frac{a}{2} \right). \end{aligned}$$

These values can be found in Abramowitz and Stegun [1], and this completes the proof.

LEMMA 9. *Let χ be a non-principal character modulo k , and let $1 < \sigma < 1.3$.*

If χ is primitive, then

$$f(0, \chi) + f(0, \bar{\chi}) < 2\kappa \log k - 0.91398.$$

If χ is imprimitive, then

$$f(0, \chi) + f(0, \bar{\chi}) < 2\kappa \log k + 0.2118.$$

Proof. If χ is primitive the lemma follows immediately from Lemmas 4 and 8. If χ is induced by the primitive character χ_1 modulo k_1 , then we have

$$\begin{aligned} f(0, \chi) + f(0, \bar{\chi}) &= f(0, \chi_1) + f(0, \bar{\chi}_1) \\ &\quad + 2 \operatorname{Re} \sum_{\substack{p|k \\ p \nmid k_1}} \chi_1(p) \log p \left(\frac{1}{\sqrt{5}(p^{\sigma_1} - 1)} - \frac{1}{p^{\sigma} - 1} \right). \end{aligned}$$

By the first part of this lemma we have

$$\begin{aligned} f(0, \chi) + f(0, \bar{\chi}) &< 2\kappa \log k_1 - 0.91398 \\ &\quad + 2 \sum_{\substack{p|k \\ p \nmid k_1}} \log p \left(\frac{1}{p^{\sigma} - 1} - \frac{1}{\sqrt{5}(p^{\sigma_1} - 1)} \right) \\ &< 2\kappa \log k - 0.91398 \\ &\quad + 2 \sum_{\substack{p|k \\ p \nmid k_1}} \log p \left(\frac{1}{p - 1} - \frac{1}{\sqrt{5}(p^{\sigma_1} - 1)} - \kappa \right). \end{aligned}$$

If $1 < \sigma < 1.3$ the terms of the sum are negative for $p > 3$, and the sum of the terms for $p = 2$ and 3 does not exceed 1.1257, so that the result follows.

LEMMA 10. Let χ be a primitive non-principal character modulo k , and $1 < \sigma < 1.3$. If χ is a complex valued character and $\rho = \beta + i\gamma$ is a zero of $L(s, \chi)$ with $|\gamma| \leq \frac{1}{2} \leq \beta$, then

$$f(0, \chi) + f(0, \bar{\chi}) < 2\kappa \log k - \frac{2(\sigma - \beta)}{(\sigma - \beta)^2 + \gamma^2} - 0.25418.$$

If χ is a real valued character and $\rho = \beta + i\gamma$ is a zero of $L(s, \chi)$ with $0 < |\gamma| \leq \frac{1}{2} \leq \beta$, then

$$f(0, \chi) < \kappa \log k - \frac{2(\sigma - \beta)}{(\sigma - \beta)^2 + \gamma^2} + 0.20281.$$

If χ is a real valued character, then

$$f(0, \chi) < \kappa \log k - \sum_{\beta} \frac{1}{\sigma - \beta} - 0.45699,$$

where the sum extends over all real zeros β of $L(s, \chi)$ with $\beta > \frac{1}{2}$.

Proof. In the case of a complex valued character, we discard all terms of the sum in Lemma 8 except for the one corresponding to the specific zero ρ . We then obtain

$$\begin{aligned} f(0, \chi) + f(0, \bar{\chi}) &< 2\kappa \log k - 0.91398 - \frac{2(\sigma - \beta)}{(\sigma - \beta)^2 + \gamma^2} \\ &\quad - 2 \operatorname{Re} \frac{1}{s - 1 + \bar{\rho}} + \frac{2}{\sqrt{5}} F(\sigma_1, \rho). \end{aligned}$$

Note that $|\gamma| \leq \frac{1}{2}$, so that

$$\begin{aligned} -\operatorname{Re} \frac{1}{s - 1 + \bar{\rho}} + \frac{1}{\sqrt{5}} F(\sigma_1, \rho) \\ < -\frac{\sigma - 1 + \beta}{(\sigma - 1 + \beta)^2 + 0.25} + \frac{1}{\sqrt{5}} F(\sigma_1, \rho). \end{aligned}$$

Since $\sigma - 1 + \beta > \frac{1}{2}$ it follows that the first term is increasing in σ and β , and also

$$F(\sigma_1, \rho) < F(\sigma_1, \beta) < F(\sigma_1, 1) < \sqrt{5}. \quad (21)$$

Thus we have

$$-\operatorname{Re} \frac{1}{s - 1 + \bar{\rho}} + \frac{1}{\sqrt{5}} F(\sigma_1, \rho) < 0.3299. \quad (22)$$

In the case of a real character χ we use Lemma 8 in the form

$$f(0, \chi) < \kappa \log k - 0.45699 - \sum'_{\beta \geq 1/2} \left[F(\sigma, \rho) - \frac{1}{\sqrt{5}} F(\sigma_1, \rho) \right].$$

Note that the zeros occur in complex conjugate pairs since χ is real. Hence for a complex zero ρ we obtain

$$\begin{aligned} f(0, \chi) &< \kappa \log k - 0.45699 - \frac{2(\sigma - \beta)}{(\sigma - \beta)^2 + \gamma^2} \\ &\quad - 2 \operatorname{Re} \frac{1}{\sigma - 1 + \bar{\rho}} + \frac{2}{\sqrt{5}} F(\sigma_1, \rho) \\ &< \kappa \log k + 0.20281 - \frac{2(\sigma - \beta)}{(\sigma - \beta)^2 + \gamma^2}. \end{aligned}$$

In order to prove the last part of the lemma we discard all terms except those corresponding to real zeros β with $\beta > \frac{1}{2}$. This yields

$$\begin{aligned} f(0, \chi) &< \kappa \log k - 0.45699 - \sum_{\beta} \frac{1}{\sigma - \beta} \\ &\quad + \sum_{\beta} \left[\frac{-1}{\sigma - 1 + \beta} + \frac{1}{\sqrt{5}} F(\sigma_1, \beta) \right]. \end{aligned}$$

The terms of the last sum are non-positive by Lemma 4, and this completes the proof.

3. PROOF OF THEOREM 2

Note that $\chi_1 \chi_2$ is non-principal since χ_1 and χ_2 are distinct, but that $\chi_1 \chi_2$ need not be primitive. For $\sigma > 1$ we have

$$\begin{aligned} &\frac{1}{\sqrt{5}} \frac{\zeta'}{\zeta}(\sigma_1) - \frac{\zeta'}{\zeta}(\sigma) + f(0, \chi_1) + f(0, \chi_2) + f(0, \chi_1 \chi_2) \\ &= \sum_{n=1}^{\infty} \frac{A(n)}{n^{\sigma}} \left(1 - \frac{1}{\sqrt{5} n^{\sigma_1 - \sigma}} \right) (1 + \chi_1(n))(1 + \chi_2(n)) \\ &\geq 0. \end{aligned} \tag{23}$$

If $1 < \sigma < 1.3$, then Lemma 3 yields

$$\frac{1}{\sqrt{5}} \frac{\zeta'}{\zeta}(\sigma_1) - \frac{\zeta'}{\zeta}(\sigma) < \frac{1}{\sigma - 1} - 0.7833. \tag{24}$$

It follows from (23), (24) and Lemmas 9 and 10 that

$$\frac{1}{\sigma - \beta_1} + \frac{1}{\sigma - \beta_2} - \frac{1}{\sigma - 1} < 2\kappa \log k_1 k_2 - 1.59 < 2\kappa \log(k_1 k_2 / 17). \quad (25)$$

We let $\sigma = 1 + r/\log M_1$, and suppose that $\min(\beta_1, \beta_2) = 1 - b/\log M_1$. From (25) we obtain

$$b > \frac{2r}{1 + 2\kappa r} - r. \quad (26)$$

We now choose $r = (\sqrt{2} - 1)/2\kappa$ in order to maximize the right hand side of (26). Since $\sigma < 1.3$, the result follows.

4. PROOF OF THEOREM 1, THE CASE $|\gamma| \geq 1$

Suppose $\rho = \beta + i\gamma$ is a zero of $L(s, \chi)$. We wish to show that $\beta < 1 - 1/(R \log M)$, where now (in a slight change of notation)

$$M = M(k, \rho) = \max\{k, k|\gamma|, 10\}.$$

Throughout the proof of the theorem, we may assume that $\gamma \geq 0$, since the zeros occur in complex conjugate pairs. Furthermore, we may assume that χ is primitive, and by Theorem 1 of Rosser and Schoenfeld [12] we may assume that χ is not the principal character.

If $\sigma > 1$, then

$$\begin{aligned} & \sum_{m=0}^4 a_m f(m\gamma, \chi^m) \\ &= \sum_{\substack{n=1 \\ (n,k)=1}}^{\infty} \frac{A(n)}{n^\sigma} \left(1 - \frac{1}{\sqrt{5} n^{\sigma_1 - \sigma}}\right) \sum_{m=0}^4 a_m \operatorname{Re} \left(\frac{\chi(n)}{n^{i\gamma}} \right)^m \\ &= \sum_{\substack{n=1 \\ (n,k)=1}}^{\infty} \frac{A(n)}{n^\sigma} \left(1 - \frac{1}{\sqrt{5} n^{\sigma_1 - \sigma}}\right) P(\arg \chi(n) - \gamma \log n) \\ &\geq 0. \end{aligned} \quad (27)$$

We now assume that $\gamma \geq 1$. Let $1 < \sigma < 1.15$. By Lemmas 3 and 5 we have

$$f(0, \chi_0) < \frac{1}{\sigma - 1} - 0.8973 - s(k), \quad (28)$$

and

$$f(\gamma, \chi) < \kappa \log k\gamma - \frac{1}{\sigma - \beta} + c(1, 1) - \kappa \log \pi. \quad (29)$$

From here the proof is separated into four cases depending on the order of χ , since the estimate of $f(m\gamma, \chi^m)$ in (27) depends on whether χ^m is principal, primitive non-principal, or imprimitive non-principal. It is sufficient in each case to show that

$$\frac{a_1}{\sigma - \beta} < \frac{a_0}{\sigma - 1} + \kappa A \log M, \quad (30)$$

for if we set $\sigma = 1 + r/\log M$ it follows that

$$\beta < 1 - \left(\frac{a_1 r}{a_0 + \kappa A r} - r \right) \frac{1}{\log M}.$$

If we choose $r = 0.33901$, then

$$\frac{a_1 r}{a_0 + \kappa A r} - r > \frac{1}{R},$$

and the result follows. Note that 0.33901 is an approximation to the optimal value of $\sqrt{a_0}(\sqrt{a_1} - \sqrt{a_0})/\kappa A$.

Consider first the case where χ is a quadratic character. It follows from (27), (28), (29) and Lemmas 6 and 7 that

$$\begin{aligned} \frac{a_1}{\sigma - \beta} &< \frac{a_0}{\sigma - 1} + \kappa A \log k\gamma - 7.652 \\ &\quad - a_0 s(k) + (a_2 + a_4)(T(k, 1) - \kappa \log k). \end{aligned} \quad (31)$$

Note that

$$\begin{aligned} &-a_0 s(k) + (a_2 + a_4)(T(k, 1) - \kappa \log k) \\ &< \sum_{p|k} \log p \left[\frac{a_2 + a_4 - a_0}{p - 1} + \frac{a_2 + a_4 + a_0}{\sqrt{5}(p^\phi - 1)} - \kappa(a_2 + a_4) \right]. \end{aligned} \quad (32)$$

For $p > 2$ the terms of this sum are negative, and the term with $p = 2$ is less than 2.179. Hence from (31) we obtain

$$\frac{a_1}{\sigma - \beta} < \frac{a_0}{\sigma - 1} + \kappa A \log k\gamma - 5.473,$$

and the result follows.

Consider next the case of a character of order 3. In this case $\chi^2 = \bar{\chi}$ and $\chi^4 = \chi$ are primitive, and it follows from (27), (28), (20) and Lemmas 6 and 7 that

$$\frac{a_1}{\sigma - \beta} < \frac{a_0}{\sigma - 1} + \kappa A \log k \gamma - 7.318 - a_0 s(k) + a_3 [T(k, 1) - \kappa \log k]. \quad (33)$$

Note that

$$\begin{aligned} & -a_0 s(k) + a_3 [T(k, 1) - \kappa \log k] \\ & < \sum_{p|k} \log p \left(\frac{a_3 - a_0}{p^{1.15} - 1} + \frac{a_3 + a_0}{\sqrt{5}(p^\phi - 1)} - \kappa a_3 \right) \\ & < 0, \end{aligned}$$

since each term in the sum is negative. The case of a cubic character then follows from (33).

If χ is of order 4, then $\chi^3 = \bar{\chi}$ is primitive and χ^2 is non-principal but need not be primitive. It follows from (27), (28), (29), and Lemmas 6 and 7 that

$$\begin{aligned} \frac{a_1}{\sigma - \beta} & < \frac{a_0}{\sigma - 1} + \kappa A \log k \gamma - 7.203 - a_0 s(k) \\ & + a_2 \left[T(k, k_2) - \kappa \log \frac{k}{k_2} \right] + a_4 [T(k, 1) - \kappa \log k], \quad (34) \end{aligned}$$

where k_2 is the conductor of χ^2 . The last three terms satisfy

$$\begin{aligned} & -a_0 s(k) + a_2 \left[T(k, k_2) - \kappa \log \frac{k}{k_2} \right] + a_4 [T(k, 1) - \kappa \log k] \\ & < \sum_{\substack{p|k \\ p \nmid k_2}} \log p \left[\frac{a_2 + a_4 - a_0}{p - 1} + \frac{a_2 + a_4 - a_0}{\sqrt{5}(p^\phi - 1)} - \kappa(a_2 + a_4) \right] \\ & + \sum_{p|k_2} \log p \left[\frac{a_4 - a_0}{p^{1.15} - 1} + \frac{a_4 + a_0}{\sqrt{5}(p^\phi - 1)} - \kappa a_4 \right]. \end{aligned}$$

In (32) we showed essentially that the first sum is less than 2.179. The terms of the second sum are negative for all p , so that the result follows from (34).

In order to complete the proof in the case $\gamma \geq 1$ we need only consider the case when χ is of order at least 5. The characters χ^2, χ^3 , and χ^4 will then be

non-principal but may be imprimitive. Let k_m denote the conductor of χ^m . It follows from (27), (28), (29), and Lemma 6 that

$$\begin{aligned} \frac{a_0}{\sigma - \beta} &< \frac{a_1}{\sigma - 1} + \kappa A \log k\gamma - 7.184 \\ &- a_0 s(k) + \sum_{m=2}^4 a_m \left[T(k, k_m) - \kappa \log \frac{k}{k_m} \right]. \end{aligned} \quad (35)$$

Note that

$$\begin{aligned} &-a_0 s(k) + \sum_{m=2}^4 a_m \left[T(k, k_m) - \kappa \log \frac{k}{k_m} \right] \\ &< -a_0 \sum_{p|k} \log p \left(\frac{1}{p^\sigma - 1} - \frac{1}{\sqrt{5}(p^{\sigma_1} - 1)} \right) \\ &\quad + \sum_{m=2}^4 a_m \sum_{\substack{p|k \\ p \nmid k_m}} \log p \left(\frac{1}{p^\sigma - 1} + \frac{1}{\sqrt{5}(p^{\sigma_1} - 1)} - \kappa \right). \end{aligned} \quad (36)$$

The terms of the last sum are negative for $p > p_\sigma$ and positive for $p \leq p_\sigma$, where p_σ is 5 or 7. Hence the condition $p \nmid k_m$ can be replaced by $p \leq p_\sigma$, and the expression on the right of (36) does not exceed

$$\sum_{\substack{p|k \\ p \leq p_\sigma}} \log p \left[\frac{a_2 + a_3 + a_4 - a_0}{p - 1} + \frac{a_2 + a_3 + a_4 + a_0}{\sqrt{5}(p^\sigma - 1)} - (a_2 + a_3 + a_4)\kappa \right],$$

and this last expression is maximized when only the primes 2 and 3 divide k , so that we obtain

$$-a_0 s(k) + \sum_{m=2}^4 a_m \left[T(k, k_m) - \kappa \log \frac{k}{k_m} \right] < 6.276. \quad (37)$$

The result then follows from (35).

5. PROOF OF THEOREM 1, THE CASE $|\gamma| < 1$, BUT $|\gamma|$ NOT TOO SMALL

The methods of Section 4 will require some modification in the case $\gamma < 1$. If any of χ^2 , χ^3 , or χ^4 is principal, then the estimate for $f(m\gamma, \chi^m)$ is affected

by the pole of $L(s, \chi_0)$ at $s = 1$. As before it suffices to prove (30) in each case.

If χ is of order at least 5, then the above difficulty does not arise, and we can proceed as before. Let $1 < \sigma < 1.15$ and let k_m be the conductor of χ^m . It follows from (27), (28), and Lemmas 5 and 6 that

$$\begin{aligned} \frac{a_1}{\sigma - \beta} &< \frac{a_0}{\sigma - 1} + \kappa A \log k - 16.746 \\ &\quad - a_0 s(k) + \sum_{m=2}^4 a_m \left[T(k, k_m) - \kappa \log \frac{k}{k_m} \right], \end{aligned}$$

and from (37) we obtain

$$\frac{a_1}{\sigma - \beta} < \frac{a_0}{\sigma - 1} + \kappa A \log M,$$

hence the result.

The cases where χ is of order 2, 3, or 4 can be treated in the same manner provided γ is bounded away from 0. Suppose first that χ is of order 2 and that $1 > \gamma \geq 0.538/\log M$. For $\sigma = 1 + 0.33901/\log M$ it follows from (27), (28), and Lemmas 5, 6, and 7 that

$$\begin{aligned} \frac{a_1}{\sigma - \beta} &< \frac{a_0}{\sigma - 1} + (a_1 + a_3) \kappa \log k + \frac{a_2(\sigma - 1)}{(\sigma - 1)^2 + 4\gamma^2} + \frac{a_4(\sigma - 1)}{(\sigma - 1)^2 + 16\gamma^2} \\ &\quad - 18.022 - a_0 s(k) + (a_2 + a_4) T(k, 1) \\ &< \frac{a_0}{\sigma - 1} + \kappa A \log M - 18.022 \\ &\quad - 0.321 \log M - a_0 s(k) + (a_2 + a_4) T(k, 1) \\ &< \frac{a_0}{\sigma - 1} + \kappa A \log M - 18.022 \\ &\quad + \sum_{p|k} \log p \left(\frac{a_2 + a_4 - a_0}{p - 1} + \frac{a_2 + a_4 + a_0}{\sqrt{5}(p^\phi - 1)} - 0.321 \right) \\ &< \frac{a_0}{\sigma - 1} + \kappa A \log M, \end{aligned} \tag{38}$$

since the terms of the sum are negative for $p > 11$.

If χ is of order 3 and $1 > \gamma \geq 0.403/\log M$, then from (27), (28), and Lemmas 5, 6, and 7 we obtain

$$\begin{aligned}
 \frac{a_1}{\sigma - \beta} &< \frac{a_0}{\sigma - 1} + \kappa A \log M - 17.609 \\
 &\quad + a_3 \left(\frac{\sigma - 1}{(\sigma - 1)^2 + 9\gamma^2} - \kappa \log M \right) + a_3 T(k, 1) - a_0 s(k) \\
 &< \frac{a_0}{\sigma - 1} + \kappa A \log M - 17.609 \\
 &\quad + \sum_{p \nmid k} \log p \left(\frac{a_3 - a_0}{p^\sigma - 1} + \frac{a_3 + a_0}{\sqrt{5}(p^{\sigma_1} - 1)} \right) \\
 &< \frac{a_0}{\sigma - 1} + \kappa A \log M,
 \end{aligned} \tag{39}$$

since the terms of the sum are negative.

Next consider the case where χ is of order 4 and $1 > \gamma \geq 0.264/\log M$. If k_2 is the conductor of χ^2 , then we obtain

$$\begin{aligned}
 \frac{a_1}{\sigma - \beta} &< \frac{a_0}{\sigma - 1} + \kappa A \log M - 16.967 - a_0 s(k) + a_4 T(k, 1) \\
 &\quad + a_4 \left(\frac{\sigma - 1}{(\sigma - 1)^2 + 16\gamma^2} - \kappa \log M \right) + a_2 \left(T(k, k_2) - \kappa \log \frac{k}{k_2} \right).
 \end{aligned} \tag{40}$$

Note that

$$\begin{aligned}
 -a_0 s(k) + a_4 T(k, 1) &\leq \sum_{p \nmid k} \log p \left(\frac{a_4 - a_0}{p^\sigma - 1} + \frac{a_4 + a_0}{\sqrt{5}(p^{\sigma_1} - 1)} \right) \\
 &< 0,
 \end{aligned} \tag{41}$$

since the terms are negative, and also that

$$\begin{aligned}
 T(k, k_2) - \kappa \log \frac{k}{k_2} &< \sum_{\substack{p \nmid k \\ p \nmid k_2}} \log p \left(\frac{1}{p - 1} + \frac{1}{\sqrt{5}(p^\sigma - 1)} - \kappa \right) \\
 &< 1.012.
 \end{aligned} \tag{42}$$

since the terms of the sum are negative for $p > 5$. It then follows from (40), (41), and (42) that

$$\frac{a_1}{\sigma - \beta} < \frac{a_0}{\sigma - 1} + \kappa A \log M - 5.15,$$

and the results follow with $\sigma = 1 + 0.33901/\log M$.

The case of a character of order 4 is unique in that it requires special consideration of zeros in the "intermediate range" $0.264 > \gamma \log M \geq 0.262$. In this case we use the inequality

$$5 + 8 \cos \theta + 4 \cos 2\theta + \cos 3\theta = (1 + 2 \cos \theta)^2(1 + \cos \theta) \geq 0, \quad (43)$$

from which it follows that

$$5f(0, \chi_0) + 8f(\gamma, \chi) + 4f(2\gamma, \chi^2) + f(3\gamma, \bar{\chi}) \geq 0. \quad (44)$$

Note that $L(\bar{\rho}, \bar{\chi}) = 0$ whenever $L(\rho, \chi) = 0$, so that from (13) we obtain

$$\begin{aligned} f(t, \bar{\chi}) = & \kappa \log \frac{k}{\pi} + \frac{1}{2} \operatorname{Re} \left[\frac{\Gamma'}{\Gamma} \left(\frac{s+a}{2} \right) - \frac{1}{\sqrt{5}} \frac{\Gamma'}{\Gamma} \left(\frac{s_1+a}{2} \right) \right] \\ & - \sum'_{\beta \geq 1/2} \left[F(s, \bar{\rho}) - \frac{1}{\sqrt{5}} F(s_1, \bar{\rho}) \right], \end{aligned}$$

where the sum extends over all zeros ρ of $L(s, \chi)$. We put $t = 3\gamma$ and discard all terms in the sum except for the one corresponding to the zero of interest. If $1 < \sigma < 1.15$ then from Lemma 1 we obtain

$$\begin{aligned} f(3\gamma, \bar{\chi}) < & \kappa \log \frac{k}{\pi} + d(1, 3) - \frac{\sigma - \beta}{(\sigma - \beta)^2 + 16\gamma^2} \\ & + \operatorname{Re} \frac{-1}{s - 1 + \bar{\rho}} + \frac{1}{\sqrt{5}} F(s_1, \bar{\rho}). \end{aligned} \quad (45)$$

Note that as in (21) we have

$$F(s_1, \bar{\rho}) < F(\sigma_1, \beta) < \sqrt{5},$$

so that (45) yields

$$f(3\gamma, \bar{\chi}) < \kappa \log k + 1.129 - \frac{\sigma - \beta}{(\sigma - \beta)^2 + 16\gamma^2}.$$

It then follows from (28), (42), (44), and Lemma 5 that

$$\frac{8}{\sigma - \beta} + \frac{\sigma - \beta}{(\sigma - \beta)^2 + 16\gamma^2} < \frac{5}{\sigma - 1} + 13\kappa \log M.$$

Let $\sigma = 1 + r/\log M$, $\beta = 1 - b/\log M$, and $\gamma = c/\log M$. We then obtain

$$\frac{8}{r + b} + \frac{r + b}{(r + b)^2 + 16c^2} < \frac{5}{r} + 13\kappa.$$

If $r = 0.33$ then the left side is decreasing in b provided $c \leq 0.264$, and it follows that $b > 0.104 > 1/R$.

6. PROOF OF THEOREM 1, THE CASE $|\gamma|$ SMALL

It remains only to consider the cases when $\gamma \log M$ is small and χ is of order 2, 3, or 4. This requires a slightly different approach followed by Gronwall [3] and Metsänkylä [7]. The use of Stechkin's device (Lemma 4) yields a numerical improvement of the results of Metsänkylä.

Suppose first that χ is of order 4 and that $\gamma \log M \leq 0.262$. We use the inequality

$$2 + 3 \cos \theta + 2 \cos 2\theta + \cos 3\theta = 4(1 + \cos \theta) \cos^2 \theta \geq 0,$$

from which it follows easily that

$$2|f(0, \chi_0) + f(0, \chi) + f(0, \bar{\chi})| + f(0, \chi^2) + f(0, \bar{\chi}^2) \geq 0.$$

If $\sigma < 1.3$ it follows from Lemmas 3, 9, and 10 that

$$\frac{2(\sigma - \beta)}{(\sigma - \beta)^2 + \gamma^2} < \frac{1}{\sigma - 1} + 3\kappa \log M.$$

We put $\sigma = 1 + r/\log M$, $\beta = 1 - b/\log M$, and $\gamma = c/\log M$. Then we have

$$\frac{2(r + b)}{(r + b)^2 + c^2} < \frac{1}{r} + 3\kappa. \quad (46)$$

Let $r = 0.62$, and note that $\sigma < 1.3$. The left side of (46) is decreasing in b and c , and if $c \leq 0.262$ it follows that $b > 0.104 > 1/R$.

If χ is of order 3 and $\gamma \log M \leq 0.403$, then we use the inequality (43), from which it follows that

$$f(0, \chi_0) + f(0, \chi) + f(0, \bar{\chi}) \geq 0.$$

If $\sigma < 1.3$ then from Lemmas 3 and 10 we obtain

$$\frac{2(\sigma - \beta)}{(\sigma - \beta)^2 + \gamma^2} < \frac{1}{\sigma - 1} + 2\kappa \log M.$$

We put $\sigma = 1 + r/\log M$, $\beta = 1 - b/\log M$, and $\gamma = c/\log M$ to get

$$\frac{2(r + b)}{(r + b)^2 + c^2} < \frac{1}{r} + 2\kappa.$$

If $r = 0.69$ and $c \leq 0.403$, then the left side is decreasing in b and c , and it follows that $b > 0.104 > 1/R$.

If χ is of order 2 and $0 < \gamma \log M = c \leq 0.538$, then we use the trivial inequality $1 + \cos \theta \geq 0$, from which it follows that

$$f(0, \chi_0) + f(0, \chi) \geq 0. \quad (47)$$

For $1 < \sigma < 1.3$ the estimates of Lemmas 3 and 10 yield

$$\frac{2(\sigma - \beta)}{(\sigma - \beta)^2 + \gamma^2} < \frac{1}{\sigma - 1} + \kappa \log M,$$

or

$$\frac{2(r + b)}{(r + b)^2 + c^2} < \frac{1}{r} + \kappa.$$

If $r = 0.69$, then $c \leq 0.538$ implies that $b > 0.104 > 1/R$.

The only case remaining is that of real zeros of L -functions formed with real non-principal characters. We first show that $L(s, \chi)$ can have at most a single zero β exceeding $1 - 1/(R \log M)$.

Suppose $\beta_1 \leq \beta_2$ are real zeros of $L(s, \chi)$ where χ is a real non-principal character. It follows from (47) and Lemmas 3 and 10 that for $1 < \sigma \leq 1.3$ we have

$$\frac{1}{\sigma - \beta_1} + \frac{1}{\sigma - \beta_2} < \frac{1}{\sigma - 1} + \kappa \log M,$$

so that

$$\frac{2}{\sigma - \beta_1} < \frac{1}{\sigma - 1} + \kappa \log M.$$

If $\sigma = 1 + r/\log M$ and $\beta_1 = 1 - b/\log M$, then we have

$$b > \frac{2r}{1 + \kappa r} - r.$$

If $M \geq 150$ we choose $r = (\sqrt{2} - 1)/\kappa$ and it follows that $b > 0.62$. If $M < 150$ then we appeal to the work of Rosser [10], where it is shown that there are no positive real zeros of L -functions associated with real non-principal characters for moduli not exceeding 227.

Finally suppose that χ_1 and χ_2 are distinct real non-principal characters modulo k with real zeros β_1 and β_2 of $L(s, \chi_1)$ and $L(s, \chi_2)$, respectively. Then from Theorem 2 we obtain

$$\min\{\beta_1, \beta_2\} < 1 - \frac{1}{R_1 \log M_1},$$

where $M_1 = \max\{k^2/17, 13\}$. From this it follows that

$$\min\{\beta_1, \beta_2\} < 1 - \frac{1}{R \log k},$$

so that at most one of β_1 or β_2 is exceptional. Hence for each k there is at most a single exceptional zero, and this completes the proof of Theorem 1.

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